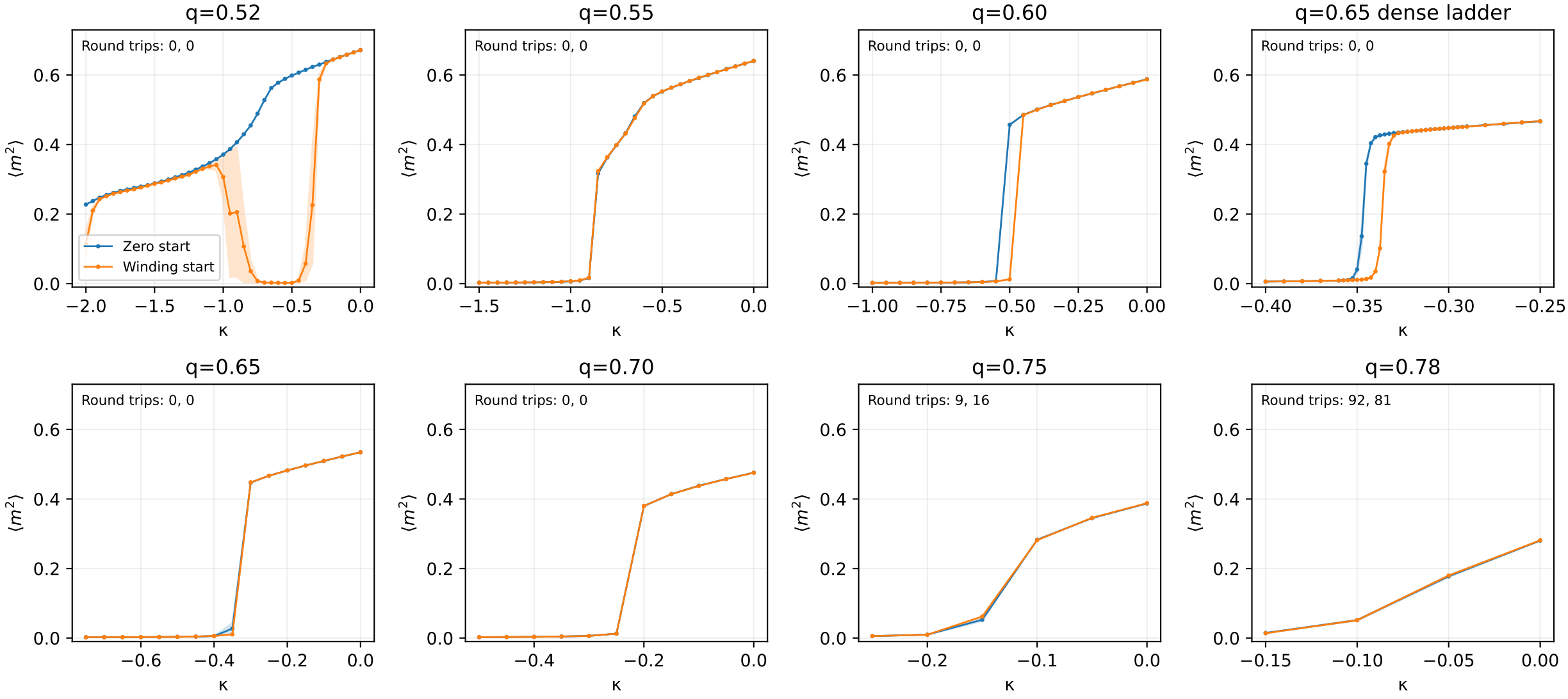
N = 8: mean uniform magnetization and two-half ranges
q=0.52
Round trips: 0, 0
Zero start
Winding start
q=0.55
Round trips: 0, 0
q=0.60
Round trips: 0, 0
q=0.65 dense ladder
Round trips: 0, 0
q=0.65
Round trips: 0, 0
q=0.70
Round trips: 0, 0
q=0.75
Round trips: 9, 16
q=0.78
Round trips: 92, 81
⟨m²⟩
κ
Each curve is one independent replica-exchange ensemble. Shading spans its two post-warmup half-means; it is not a confidence band.
Agreement of these diagnostics does not establish equilibrium; unresolved transport and metastability remain explicit.